

Lab OS Network tools

Command: ipconfig

- Displays all current TCP/IP network configuration values
- i.e.: **ipconfig /all**

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Windows 2000 IP Configuration

Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

   Host Name . . . . . : host.grape-info.com
   DNS Servers . . . . . : 10.1.0.200
   Description . . . . . : 3Com 3C90X Ethernet Adapter
   Physical Address. . . . . : 00-60-08-3E-46-07
   DHCP Enabled. . . . . : Yes
   Autoconfiguration Enabled . . . . . : Yes
   IP Address. . . . . : 192.168.0.112
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 192.168.0.2
   DHCP Server . . . . . : 192.168.0.1
   Primary WINS Server . . . . . : 192.168.0.1
   Secondary WINS Server . . . . . : 192.168.0.3
   Ob   i   d           d   d   S       b   02  1998  10  32  13
```

PS: Get-NetIPConfiguration -Detailed

Command: ipconfig

- If the computer is DHCP-enabled and is using a DHCP server to obtain configuration, you can initiate a refresh of the lease by using the **ipconfig /renew** command
- When you use **ipconfig /renew**, all network adapters on the computer that uses DHCP renew their existing configuration
- It's possible to use the **ipconfig** command with the **/release** option to immediately release the current DHCP configuration

Command: ipconfig

- You can also use the **ipconfig** command to:
 - Display or reset the DNS cache (/displaydns - /flushdns).
 - Refresh registered DNS names (/registerdns).
 - Display the DHCP class IDs for an adapter (/showclassid).
 - Set the DHCP class IDs for an adapter (/setclassid).

Command: ping

- The **ping** command helps to verify IP-level connectivity.
- Use **ping** whenever you need to verify that a host computer can connect to the TCP/IP network and network resources.
- i.e.: **ping 192.168.1.10**

Command: ping

- You should perform the following steps when using **ping**:
 - Ping the loopback address to verify that TCP/IP is installed and configured correctly on the local computer
 - **ping 127.0.0.1**
 - Ping the IP address of the local computer to verify that it was added to the network correctly.
 - **ping IP_address_of_local_host**
 - Ping the IP address of the default gateway to verify that the default gateway is functioning and that you can communicate with a local host on the local network.
 - **ping IP_address_of_default_gateway**
 - Ping the IP address of a remote host to verify that you can communicate through a router.
 - **ping IP_address_of_remote_host**

Command: ping

- The **ping** command uses Windows Sockets—style name resolution to resolve a computer name to an IP address
 - If pinging by address succeeds, but pinging by name fails, then the problem lies in address or name resolution, not network connectivity

Command: ping

- It's possible to use different options with the **ping** command to specify
 - the size of packets to use (-l)
 - how many packets to send (-n)
 - what Time-to-Live (TTL) value to use (-i)
 - whether to set the "don't fragment" flag. (-f)
- By default, **ping** waits 1,000 ms (1 second) for each response to be returned before displaying the "Request Timed Out" message.
- PS: **Test-NetConnection**

Command: tracert

- Find the IP address of any remote host.
- TRACERT is useful for troubleshooting large networks where several paths can be taken to arrive at the same point, or where many intermediate systems (routers or bridges) are involved.
- i.e.: **tracert** www.google.it

Command: tracert

- **Tracert** uses the IP TTL field and ICMP error messages to determine the route from one host to another through a network
- Care must be taken with tracert as it shows the optimal route, not necessarily the actual route

Command: tracert

- **Tracert** determines the path by sending the first Echo Request message with a TTL of 1 and incrementing the TTL by 1 on each subsequent transmission until either the target host responds or the maximum number of hops is reached.
- TTL is effectively a count of the (maximum) number of links to the destination host.
- Each router along the path decrements the TTL in an IP packet by at least 1 before forwarding it.

Command: pathping

- Combines the functionality of ping and tracert
- Trace route and provide network latency and packet loss for each router and link in the path
- **Pathping** is invaluable for determining which routers or subnets may be having network problems
- It displays the degree of packet loss at any given router or link.

Command: pathping

- Pathping performs the equivalent of the **tracert** command by identifying which routers are on the path.
- When **pathping** is run, you first see the results for the route as it is tested for problems.
- The **pathping** command then displays a busy message for the next xxx seconds (this time varies by the hop count). During this time, **pathping** gathers information from all the routers previously listed and from the links between them.
- The two rightmost columns—**This Node/Link Lost/Sent=Pct** and **Address**—contain the most useful information.

Command: netstat

- You can use the **netstat** command to display protocol statistics and current TCP/IP connections.
 - **netstat -a** command displays all connections
 - **netstat -r** displays the route table plus active connections
 - **netstat -e** command displays Ethernet statistics
 - **netstat -s** displays per-protocol statistics
 - **netstat -n**, addresses and port numbers are not converted to names
- PS: **Get-NetTCPConnection**

Command: nbtstat

- NetBIOS over TCP/IP (NetBT) resolves NetBIOS names to IP addresses
 - **nbtstat -n** displays the names that were registered locally on the system by programs such as the server and redirector.
 - **nbtstat -c** shows the NetBIOS name cache, which contains name-to-address mappings for other computers.
 - **nbtstat -R** purges the name cache and reloads it from the Lmhosts file.
 - **nbtstat -RR** releases NetBIOS names registered with a WINS server and then renews their registration.
 - **nbtstat -a *name*** performs a NetBIOS adapter status command against the computer specified by *name*. The adapter status command returns the local NetBIOS name table for that computer plus the media access control address of the adapter.
 - **nbtstat -S** lists the current NetBIOS sessions and their status, including statistics

Command: arp

- The Address Resolution Protocol (ARP) allows a host to find the media access control address of a host on the same physical network, given the IP address of the host.
- To make ARP efficient, each computer caches IP-to-media access control address mappings to eliminate repetitive ARP broadcast requests.
- You can use the **arp** command to view and modify the ARP table entries on the local computer.

Various Commands

- **Nslookup**: displays information about DNS records for specific IP addresses
- **Getmac**: Discovers MAC address and lists associated network protocols for all NIC in a PC
- **Hostname**: display the hostname of the current computer
- **Route**: displays information about route of the packets
- **Netsh**: Network shell (netsh) is a command-line utility that allows you to configure and display the status of various network communications server roles and components installed on computers running Windows
 - Es: **netsh wlan show interfaces**